

Conditional Wasserstein generative adversarial network-gradient penalty-based approach to alleviating imbalanced data classification



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ARTICLE INFO

Article history:

Received 29 March 2019

Revised 6 October 2019

Accepted 12 October 2019

Available online 13 October 2019

Keywords:

Imbalanced learning

Oversampling approach

SMOTE

GAN

WGAN-GP

ABSTRACT

In data mining, common classification algorithms cannot effectively learn from imbalanced data. Oversampling addresses this problem by creating data for the minority class in order to balance the class distribution before the model is trained. The Traditional oversampling approaches are based on Synthetic Minority Oversampling Technique (SMOTE), which focus on local information but generates insufficiently realistic data. In contrast, the Generative Adversarial Network (GAN) captures the true data distribution in order to generate data for the minority class. However, both approaches are problematic owing to mode collapse and unstable training. To overcome these problems, we propose Conditional Wasserstein GAN- Gradient Penalty (CWGAN-GP), a novel and efficient synthetic oversampling approach for imbalanced datasets, which can be constructed by adding auxiliary conditional information to the WGAN-GP. CWGAN-GP generates more realistic data and overcomes the aforementioned problems. Experiments on 15 different benchmarked datasets and two real imbalanced datasets empirically demonstrate that CWGAN-GP increases the quality of synthetic data; furthermore, our approach outperforms the other oversampling approaches based on three evaluation metrics (F-measure, G-mean, and the area under the receiver operating characteristic curve) for five classifiers. Friedman and Nemenyi post hoc statistical tests also confirm that CWGAN-GP is superior to the other oversampling approaches.

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1. Introduction

In the fields of machine learning and data mining, imbalanced data classification is a ubiquitous natural phenomenon. Imbalanced data classification generally refers to the imbalanced distribution of samples among different categories of a dataset. In binary or multiclass datasets affected by imbalanced data classification, a category with fewer samples is referred to as a positive class or a minority class, whereas a category with more samples is known as a negative class or a majority

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class. In practice, imbalanced data classification is associated with text classification [1], consumer purchasing decisions based on behavioral history [2], fraud detection based on credit card transactional data [3,4], and disease diagnosis based on medical data [5,6]. As traditional classification algorithms are based on situations in which the data distribution is relatively balanced, these algorithms may experience different degrees of defects when faced with imbalanced data classification and may thus become inefficient [7]. Therefore, enhancing the performance of the classifier in the case of imbalanced data and increasing the ability to identify the minority class are extremely important problems that require urgent solutions.

Current research on the classification of imbalanced data is mainly concerned with two methods [8]. The first involves newly designing or improving the classification algorithm, which generally entails reducing the sensitivity of the classification algorithm to the imbalanced data. Typically, either ensemble learning is used to increase the robustness of the classification algorithm [9–11] or a cost-sensitive learning method [4,12] is used to make the cost of misclassifying the minority class higher than that of misclassifying the majority class. The second approach is to use a sampling method to balance the dataset on the data level, mainly via oversampling [13], undersampling [14], or hybrid sampling [15]. The purpose of oversampling is to increase the number of samples in the minority class; thus, improving the distribution of data among classes. Undersampling has the same purpose as oversampling, but instead removes samples from the majority class. Finally, hybrid sampling is intended to balance the dataset by combining the two aforementioned sampling methods.

This study considers the oversampling approach in binary imbalanced data learning; this focus was chosen for two reasons. First, multi-classification problems can usually be solved by simplifying multiple binary classification issues by using One-Against-All ideas [16]. Therefore, research to address the problems associated with imbalanced data learning typically focus on improving the overall classification performance of binary imbalanced data, which is the focus of this study. Second, because the oversampling approach is independent of the selected classifier, imbalanced data only need to be oversampled once to train different classifiers; thus, the advantages of using the oversampling approach are of a more general nature.

Current oversampling methods mainly belong to the following two categories: traditional oversampling methods and methods based on a Generative Adversarial Network (GAN). Most traditional oversampling methods are based on the Synthetic Minority Oversampling TEchnique (SMOTE) [13,17], and they mainly generate synthetic samples along the line segment that joins minority class samples. However, because these methods are more concerned with local data information, the data they generate may not be sufficiently realistic. The second category of oversampling methods is based on GAN [18], which can capture the true distribution of data and directly generate synthetic data. Douzas et al. [19] introduced a network based on the Conditional Generative Adversarial Network (CGAN) [20] that captures the true distribution of the global minority class by adding additional conditional information to GAN. Then, in the case of imbalanced data, it generates synthetic data for the minority class; however, this suffers from unstable training and mode collapse [19,21–23].

This study aims to thoroughly address the above problems by proposing the CWGAN-GP model, which combines WGAN-GP [24] with auxiliary conditional information as a new oversampling approach to generate synthetic samples for the minority class in an imbalanced dataset. CWGAN-GP not only generates more realistic data but also overcomes the problems of mode collapse and unstable training. Our three main reasons for proposing the CWGAN-GP oversampling method are as follows. First, rather than being based on the local information of imbalanced datasets, CWGAN-GP is based on the global information of the real data distribution; as such, that the generated data is more realistic. Second, both the GAN and CGAN models are problematic in that they are unstable during training and undergo mode collapse, whereas CWGAN-GP rarely experiences these problems. Third, WGAN does not thoroughly consider the additional conditional information; in contrast, CWGAN-GP not only offers some of the advantages of WGAN but also draws on the advantages of the auxiliary conditional information of CGAN, and thus increases the quality of generative synthetic data [25].

The main contributions of this study can be summarized as follows.

- (1) We propose CWGAN-GP as a new oversampling approach that can generate nearly real minority samples based on the real distribution of data to increase the low recognition rate of the minority class in an imbalanced dataset.
- (2) The data generated by traditional oversampling approaches based on the local information of a dataset may not be sufficiently realistic; to address this problem, the proposed CWGAN-GP oversampling approach proposed can learn the real data distribution based on the global information of a dataset to generate more realistic data.
- (3) GAN-based oversampling approaches exhibit mode collapse and training instability; the proposed CWGAN-GP oversampling approach mostly refrains from this behavior.

The remainder of this paper is organized as follows. Section 2 describes the proposed approach in detail. Section 3 presents the experimental methodology and performance evaluation measures used to validate our oversampling approach. The experimental results are discussed in Section 4, and related work is discussed in Section 5. Finally, Section 6 briefly summarizes our study and presents the conclusions.

2. Proposed methods

This section presents our analysis of GAN and its conditional version. We then introduce the Conditional WGAN-GP architecture, which combines WGAN-GP and the auxiliary conditional information. Specifically, GAN and CGAN are introduced in Sections 2.1, and 2.2 presents WGAN and WGAN-GP. Finally, Section 2.3 proposes CWGAN-GP and describes the addition of auxiliary conditional information to the WGAN-GP.

2.1. GAN and CGAN

The GAN training procedure consists of defining a game between two adversarial networks: a generator network G maps a source of noise to the input space, and a discriminator network D distinguishes whether a sample originated from the true data distribution or from the generative data distribution. The D and G play the following two-player minimax game with Eq. (1):

$$\min_G \max_D V(D, G) = \mathbb{E}_{\mathbf{x} \sim P_r} [\log D(\mathbf{x})] + \mathbb{E}_{\mathbf{x} \sim P_g} [\log(1 - D(\mathbf{x}))] \quad (1)$$

where P_r is the true data distribution and P_g is the generative data distribution implicitly defined by $\mathbf{x} = G(\mathbf{z})$, $\mathbf{z} \sim p(\mathbf{z})$, for which $\mathbf{z}$ is sampled from a simple noise distribution p , such as a uniform, normal, or Gaussian distribution. The D is optimized to maximize the probability that the training samples and samples from G are correctly classified, and G is optimized to minimize $\mathbb{E}_{\mathbf{x} \sim P_g} [\log(1 - D(\mathbf{x}))]$ or $\mathbb{E}_{\mathbf{x} \sim P_g} [-\log D(\mathbf{x})]$ [18].

The CGAN, which is a conditional version of the GAN framework, adds additional information $\mathbf{y}$ to both the generator and discriminator. The information $\mathbf{y}$ could either be the class label or any other type of auxiliary information. In addition, the training procedure for CGAN is identical to that of GAN.

Formally, the objective value function between the generator G and the discriminator D is the minimax Eq. (2):

$$\min_G \max_D V(D, G) = \mathbb{E}_{\mathbf{x} \sim P_r} [\log D(\mathbf{x}|\mathbf{y})] + \mathbb{E}_{\mathbf{x} \sim P_g} [\log(1 - D(\mathbf{x}|\mathbf{y}))] \quad (2)$$

where P_r and P_g have the same meaning as those in GAN, and $\mathbf{y}$ is combined with prior noise as inputs to the hidden layer. The optimization of CGAN is similar to that of GAN.

As CGAN is an extension of the GAN model, CGAN exhibits the same problematic behavior; namely, mode collapse and unstable training (vanishing gradients), among other issues. Furthermore, researchers have pointed out that when GAN and CGAN use the Jensen–Shannon (JS) divergence as a measure of generative samples, the model would be able to generate only continuous data rather than discrete data [21]:

$$JS(P_r, P_g) = KL(P_r || P_m) + KL(P_g || P_m) \quad (3)$$

where KL is the Kullback–Leibler divergence and P_m is the mixture $(P_r + P_g)/2$.

2.2. WGAN and WGAN-GP

Unlike GAN and CGAN, WGAN and WGAN-GP use the *Earth-Mover* (EM) distance (also known as Wasserstein-1) rather than the JS divergence as a measure of the distance between the true data distribution and the generative data distribution, because the smoothness of the EM distance is superior to that of the JS divergence. On the one hand, WGAN can theoretically solve the problem of vanishing gradients in GAN and CGAN; on the other hand, previous experiments have revealed that WGAN can effectively address the problem of mode collapse by replacing the JS divergence with the EM distance [21,22]:

$$W(P_r, P_g) = \inf_{\gamma \in \Pi(P_r, P_g)} \mathbb{E}_{(x,y) \sim \gamma} [|x - y|] \quad (4)$$

where $\Pi(P_r, P_g)$ denotes the entire set of feasible joint distributions $\gamma(x, y)$ of the true data distribution P_r and generative data distribution P_g . The $W(P_r, P_g)$ is formally defined as the minimum cost of transporting mass to transform distribution P_r into distribution P_g . Furthermore, under mild assumptions, $W(P_r, P_g)$ is continuous and differentiable almost everywhere. However, Eq. (4) is highly intractable; thus, the EM distance could be reconstructed using the Kantorovich–Rubinstein duality [26]:

$$W(P_r, P_g) = \frac{1}{K} \sup_{\|f\|_L \leq K} \mathbb{E}_{\mathbf{x} \sim P_r} [f(\mathbf{x})] - \mathbb{E}_{\mathbf{x} \sim P_g} [f(\mathbf{x})] \quad (5)$$

where the supremum is over all the K -Lipschitz functions $f: \mathbf{x} \rightarrow \mathbb{R}$, and K denotes K -Lipschitz for constant K , where $K=1$ for the original WGAN. The objective value function between the generator and the discriminator (known as the *critic* in WGAN) is the minimax Eq. (6):

$$\min_G \max_D V(D, G) = \mathbb{E}_{\mathbf{x} \sim P_r} [D(\mathbf{x})] - \mathbb{E}_{\mathbf{x} \sim P_g} [D(\mathbf{x})] \quad (6)$$

However, the WGAN still tends to generate only poor samples or fails to converge. Gulrajani et al. [24] found that these problems are often due to weight clipping in WGAN, which is clearly a highly ineffective way to enforce a Lipschitz constraint on the discriminator. Thus, they proposed an alternative to clipping weights by penalizing the norm of the gradient of the discriminator with respect to its input; this approach is an extension of WGAN and is known as WGAN-GP. Their experimental results revealed that WGAN-GP outperforms standard WGAN and enables the stable training of a wide variety of GAN architectures with almost no hyperparameter tuning. Formally, the new object value function of WGAN-GP is the minimax Eq. (7):

$$\min_G \max_D V(D, G) = \mathbb{E}_{\mathbf{x} \sim P_r} [D(\mathbf{x})] - \mathbb{E}_{\tilde{\mathbf{x}} \sim P_g} [D(\tilde{\mathbf{x}})] - \lambda \mathbb{E}_{\tilde{\mathbf{x}} \sim P_g} [(\|\nabla_{\tilde{\mathbf{x}}} D(\tilde{\mathbf{x}})\|_2 - 1)^2] \quad (7)$$

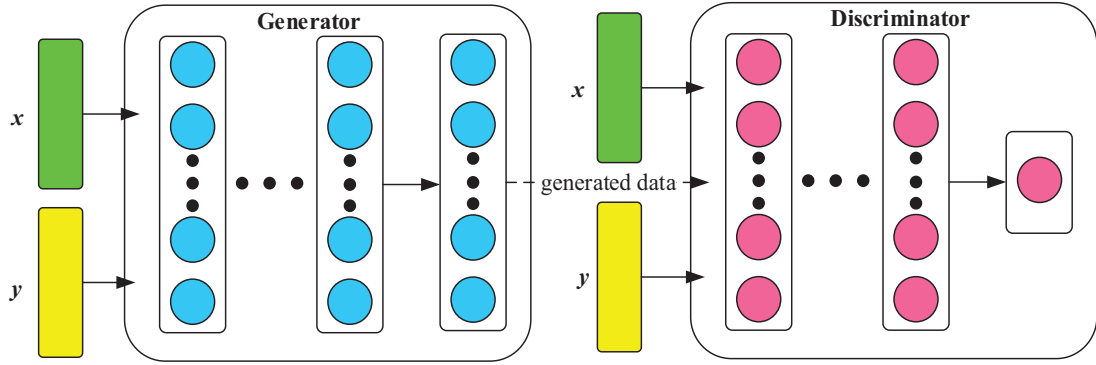


Fig. 1. Structure of CWGAN-GP.

where λ is the gradient penalty coefficient and $\hat{\mathbf{x}}$ is sampling along straight lines between the true data distribution P_r and the generative data distribution P_g :

$$\hat{\mathbf{x}} = \varepsilon \mathbf{x} + (1 - \varepsilon) \tilde{\mathbf{x}}, \varepsilon \sim \text{Uniform}[0, 1], \mathbf{x} \sim P_r, \tilde{\mathbf{x}} \sim P_g \quad (8)$$

2.3. CWGAN-GP

The motivation for the development of WGAN-GP can be extended to CWGAN-GP if both the discriminator and generator are conditioned on some auxiliary additional information $\mathbf{y}$ originating from the CGAN version of GAN. Here, $\mathbf{y}$ could be any type of information; in this study, $\mathbf{y}$ is the class label. In the discriminator, we concatenate both P_r and P_g with $\mathbf{y}$ in joint hidden representation; in the generator, we also concatenate $\mathbf{y}$ with $p(\mathbf{z})$ in the same representation. Formally, the objective value function between the generator and the discriminator is the minimax Eq. (9):

$$\min_G \max_D V(D, G) = \mathbb{E}_{\mathbf{x} \sim P_r} [D(\mathbf{x}|\mathbf{y})] - \mathbb{E}_{\tilde{\mathbf{x}} \sim P_g} [D(\tilde{\mathbf{x}}|\mathbf{y})] - \lambda \mathbb{E}_{\hat{\mathbf{x}} \sim P_{\hat{\mathbf{x}}}} [(\|\nabla_{\hat{\mathbf{x}}} D(\hat{\mathbf{x}}|\mathbf{y})\|_2 - 1)^2] \quad (9)$$

where λ is once again the gradient penalty coefficient and the sampling mode of $\hat{\mathbf{x}}$ is the same as that for WGAN-GP.

The loss functions of the discriminator and generator are minimized as follows, and the structure of CWGAN-GP is shown in Fig. 1:

$$L(D) = -\mathbb{E}_{\mathbf{x} \sim P_r} [D(\mathbf{x}|\mathbf{y})] + \mathbb{E}_{\tilde{\mathbf{x}} \sim P_g} [D(\tilde{\mathbf{x}}|\mathbf{y})] + \lambda \mathbb{E}_{\hat{\mathbf{x}} \sim P_{\hat{\mathbf{x}}}} [(\|\nabla_{\hat{\mathbf{x}}} D(\hat{\mathbf{x}}|\mathbf{y})\|_2 - 1)^2] \quad (10)$$

$$L(G) = -\mathbb{E}_{\tilde{\mathbf{x}} \sim P_g} [D(\tilde{\mathbf{x}}|\mathbf{y})] \quad (11)$$

3. Experimental methodology

This study aims to evaluate the performance of the data generator of CWGAN-GP as an oversampling approach to solve the binary classification problem of imbalanced data, primarily by enhancing the classification performance of imbalanced data. The performance of the CWGAN-GP generator as an oversampling approach is evaluated and compared with that of the traditional methods: SMOTE [13], BORDERLINE-SMOTE [27], Adaptive Synthetic (ADASYN) [28], Random OverSampling (ROS), without sampling (NONE), and methods based on GAN and its extended versions (i.e., GAN, CGAN, and WGAN-GP).

First, we introduce the benchmark datasets in Section 3.1. Next, we briefly describe the parameter settings of the baseline methods of the oversampling approaches and WCWGAN-GP. Finally, Section 3.2 briefly explains the parameter settings of classification algorithms that have been used in similar classification studies of class imbalanced data [9,19,29] to evaluate the effects of each sampling approach on the benchmark datasets. Then, Section 3.3 explains the experimental design and the metrics used to evaluate performance. Finally, in Section 3.4, we present the statistical tests that were employed to compare the results of the classifiers.

3.1. Benchmark dataset

In this study, we experimented on 15 different benchmarked datasets extracted from the UCI Machine Learning Repository [30,31]. The effectiveness of our approach was experimentally verified using two real imbalanced datasets from different fields. The first of these datasets contains traffic accident data for an 8-year period (2009–2016) obtained from the Leeds

Table 1

Characteristics of the imbalanced datasets used in the experimental study.

ID	Dataset	#Instances	#Features	Minority class	Majority class	#Minority instances	#Majority instances	IR
1	Heart	270	12	2	1	120	150	1.25
2	Liver	345	6	1	2	145	200	1.38
3	Lymphography	148	18	3	all other	61	87	1.43
4	Iris	150	4	versicolor	all other	50	100	2.00
5	Wine	178	13	1	all other	59	119	2.02
6	German	1000	24	2	1	300	700	2.33
7	Lung	27	56	1	all other	8	19	2.38
8	Haberman	306	3	2	1	81	225	2.78
9	Vehicle	846	17	bus	all other	218	628	2.88
10	Hepatitis	80	19	1	2	13	67	5.15
11	Ecoli	336	7	pp	all other	52	284	5.46
12	Balance	625	4	B	all other	49	576	11.76
13	Dermatology	358	34	6	all other	20	338	16.90
14	Yeast	1484	8	VAC	all other	30	1454	48.47
15	Pageblocks	5473	10	3	all other	28	5445	194.46

Table 2

Characteristics of the real imbalanced datasets used in the experimental study.

ID	Dataset	#Instances	#Features	Minority class	Majority class	#Minority instances	#Majority instances	IR
1	Traffic accident	18727	12	Fatal	Non-Fatal	132	18595	141
2	Software defect	705	37	Defect	Non-Defect	61	644	11

Table 3

Parameter settings of traditional oversampling approaches based on SMOTE.

Oversampling approaches	k_neighbors	m_neighbors	n_neighbors	return_indices
SMOTE	5	Deprecated	None	None
Borderline SMOTE	5	10	None	None
ADASYN	None	None	5	None
ROS	None	None	None	False

Traditional oversampling approaches based on SMOTE were implemented based on Imbalanced-learn [33].

City Council, United Kingdom [32]; the second consists of public software defects provided by the National Aeronautics and Space Administration (NASA). These datasets are publicly available on the corresponding web pages.^{1,2,3}

3.1.1. Machine learning repository UCI datasets

Multiclass datasets in the UCI dataset were transformed into binary datasets by combining one or more classes, which were labeled as the majority class to obtain binary datasets with different Imbalance Ratios (IRs). The IR is defined as the ratio between the majority and minority classes. Table 1 summarizes the 15 datasets, including the total number of instances, numbers of features, class names belonging to the minority and majority class, numbers of instances belonging to these classes, and the IR.

3.1.2. Real imbalanced datasets

After incomplete, erroneous, and repeated data were deleted, the traffic accident dataset comprised 18,727 accident records. Of these records, 18,595 were non-fatal traffic accidents and the remaining 132 records were fatal traffic accidents. The IR of this dataset is 141, which indicates a typical imbalanced dataset. The PC1 dataset (NASA) contains 705 records of software defects, of which 644 records are non-defective data and the remaining 61 records are defective data. In this case, the defect rate is 8.65 and the IR is 11, which also indicates a typical imbalanced dataset. The datasets are summarized in Table 2.

3.2. Parameter settings

The performance of CWGAN-GP was compared with those of traditional oversampling approaches based on SMOTE and generation approaches based on GAN; thus, the parameter settings of the baseline approaches focused on two aspects. The parameter settings of traditional oversampling approaches based on SMOTE are listed in Table 3.

With respect to the generation approaches based on GAN, most hyperparameters were the same, except for the loss function, the activation function, and the number of D parameters that were updated, followed by the update of a single

¹ <http://archive.ics.uci.edu/ml/index.php>.

² <https://data.gov.uk/dataset/6efe5505-941f-45bf-b576-4c1e09b579a1/road-traffic-accidents>.

³ <http://openscience.us/repo/>.

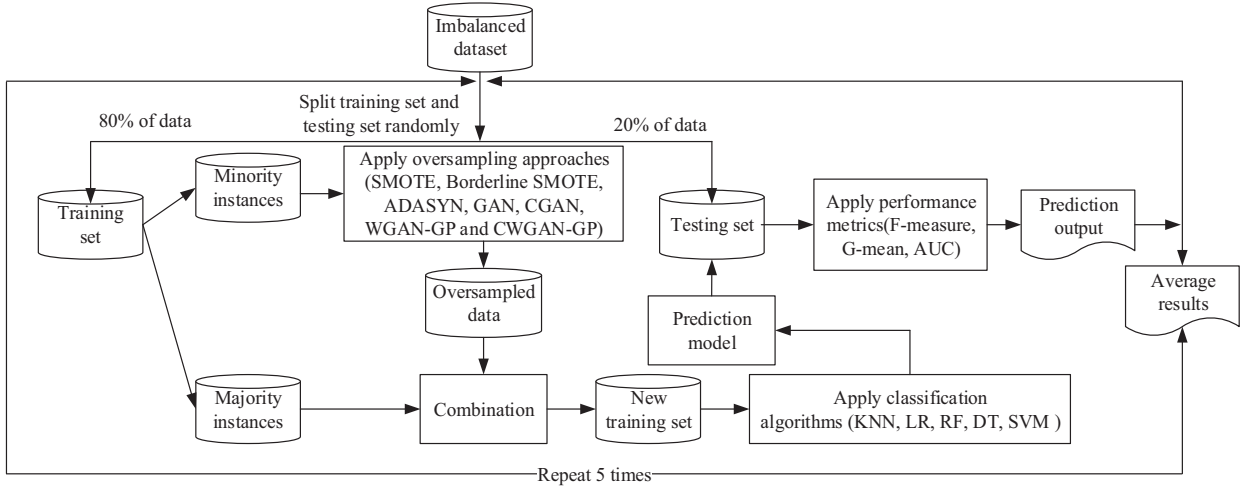


Fig. 2. Flowchart showing the experimental steps for each dataset.

G parameter. With respect to the same hyperparameters, GAN, CAGN, WGAN-GP, and CWGAN-GP were all four-layer neural networks, the dimension of noise space was set to range from 25 to 100, and neither Dropout nor Batch Normalization were applied to either the generator or the discriminator in each model. The numbers of units for the hidden layers of the generator and discriminator were set to 32–256 and 16–128, respectively. The batch size was set to 32 and the number of epochs was 5000. From the perspective of different hyperparameters, the loss functions of GAN, CGAN, and WGAN-GP were consistent with those originally reported [18,20,24]. Both the generator and discriminator used rectified linear units [34] as the activation function for the hidden layers and input layer. The output layer of the discriminator in GAN and CGAN used sigmoid activation, but no activation was applied to the output layer in either WGAN-GP or CWGAN-GP. GAN and CGAN were trained using the default settings of the Adam optimizer [35]. Although WGAN-GP and CWGAN-GP were also trained using this optimizer, the values of α , β_1 , and β_2 of the Adam optimizer were set to 0.0001, 0, and 0.9, respectively. Similarly, unlike GAN and CGAN, the gradient penalty coefficient λ in WGAN-GP and CWGAN-GP was set to 10. In GAN and CGAN, one D parameter was updated followed by the single G parameter; in WGAN-GP and CWGAN-GP, five were updated. The data generation approaches based on GAN were implemented in Python using Google's open-source framework TensorFlow [36].

Additionally, k-Nearest Neighbor (KNN) [37], Logistic Regression (LR) [38], Random Forest (RF) [39], Decision Tree (DT) [40], and Support Vector Machine (SVM) with a radial basis function [41] were used as classification algorithms to evaluate the above oversampling approaches. These five algorithms were implemented based on the Python library Scikit-Learn [42] with default settings employed.

3.3. Experimental design and performance evaluation metrics

The five above-mentioned classification algorithms (Section 3.2) were used to experimentally evaluate the performance of the eight oversampling approaches and the approach without sampling on each dataset. To prevent the testing set from being affected by synthetic data, it was isolated from the training set during preprocessing; hence, we only subjected the training set to oversampling. We randomly selected 20% of the sample set as the testing set, and we applied oversampling to the remaining 80% of the dataset as the training set. Fig. 2 shows the flowchart for empirically evaluating oversampling approaches for each dataset. The experiment was repeated five times to reduce the impact of bias caused by randomly partitioning the training and testing sets.

We used the F-measure, G-mean, and receiver operating characteristic (ROC) area under the curve (AUC) as performance metrics for evaluating the effect of each oversampling approach on the datasets; all of these metrics have been used in similar class imbalance studies [19,29,43–45]. The performance evaluation metrics are defined as follows.

$$F - measure = \frac{(1 + \beta^2) \cdot \frac{TP}{TP+FN} \cdot \frac{TP}{TP+FP}}{\beta^2 \cdot \frac{TP}{TP+FN} + \frac{TP}{TP+FP}} \quad (12)$$

where β is usually equal to 1, as is the case in this study; TP (true positive) represents a real case; that is, the real category is a true positive, and the predicted class is a true positive; FP (false positive) represents a false positive; that is, the real category is a negative and the predicted class is a positive; and FN (false negative) represents a false negative case; that is, the real category is a positive, and the predicted class is a negative.

$$G - mean = \sqrt{\frac{TP}{TP+FN} \times \frac{TN}{TN+FP}} \quad (13)$$

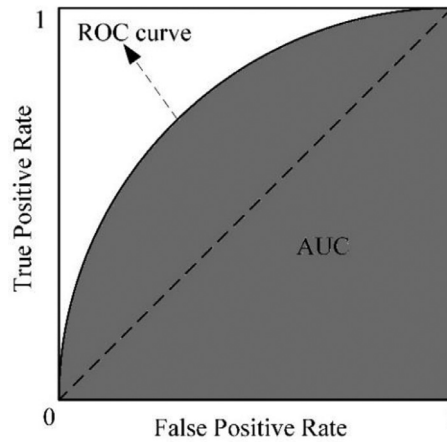


Fig. 3. ROC curve and AUC diagram.

where TN (true negative) represents a true negative case; that is, the real category is false negative, and the predicted class is a negative.

The larger the AUC, the more effective the classifier. Fig. 3 illustrates the calculation of the AUC on a two-dimensional chart, where the gray area is the AUC value.

In Fig. 3, the x-axis represents the False Positive Rate (FPR); that is, the proportion of misclassified negative cases relative to the total number of negative cases. The y-axis represents the True Positive Rate (TPR); that is, the proportion of correctly predicted positive cases relative to the total number of positive cases. As the ROC curve depends on the classification threshold, the AUC is a useful performance evaluation metric of the classifier because it is independent of the decision criterion [45].

3.4. Statistical tests

We employed non-parametric testing to analyze and compare whether significant differences exist between the oversampling approaches on multiple datasets. These tests have been used in several empirical studies and are highly recommended in the field of machine learning [9,46] to evaluate experimental results for the F-measure, G-mean, and AUC. The non-parametric test procedure consists of three steps. First, ranking scores are computed by assigning a rank score to each oversampling approach in every dataset and metric. Thus, a 9 is assigned to the best approach, 8 to the second-best approach, and so on. The ranking score of the worst approach in this study is 1 because we consider a situation in which sampling is not used. Next, the Friedman test [47] is used to determine whether these oversampling approaches deliver the same performance. If performances differ, the hypothesis that all oversampling approaches have the same performance is rejected; if the performance of the oversampling approach is significantly different, a post hoc test is needed to further distinguish each oversampling approach. Finally, when the hypothesis that all oversampling approaches have the same performance is rejected, the Nemenyi post hoc test [48] is applied to check whether the control oversampling approach (usually the best one) significantly outperforms the remaining oversampling approaches. The Nemenyi post hoc procedures enable calculating the critical distance of the mean ranking score difference. If the difference between the mean ranking scores of the two oversampling approaches exceeds the critical distance, the hypothesis that the performance of the two oversampling approaches is the same is rejected at a specified level of significance α (i.e., there exist significant differences); in this study, $\alpha = 0.05$.

4. Results and discussion

This section presents the experimental results of the oversampling approaches for different classification algorithms. The performance of CWGAN-GP is compared with those of SMOTE, BORDERLINE-SMOTE, ADASYN, ROS, GAN, CAN, WGAN-GP, and no sampling (NONE) from two perspectives (Sections 4.1 and 4.2). Section 4.3 presents the performance evaluation results for two real imbalanced datasets, and in Section 4.4, we discuss three aspects of the oversampling approaches and the experiment.

4.1. Performance evaluation results on UCI datasets

In Fig. 4, each plot corresponds to a specific classifier and evaluation metric. The x- and y-axes of each plot represent the different imbalanced datasets with different IRs and the score of the evaluation metric, respectively. Therefore, each plot includes nine lines, corresponding to eight oversampling approaches and the approach without oversampling. Fig. 4 shows

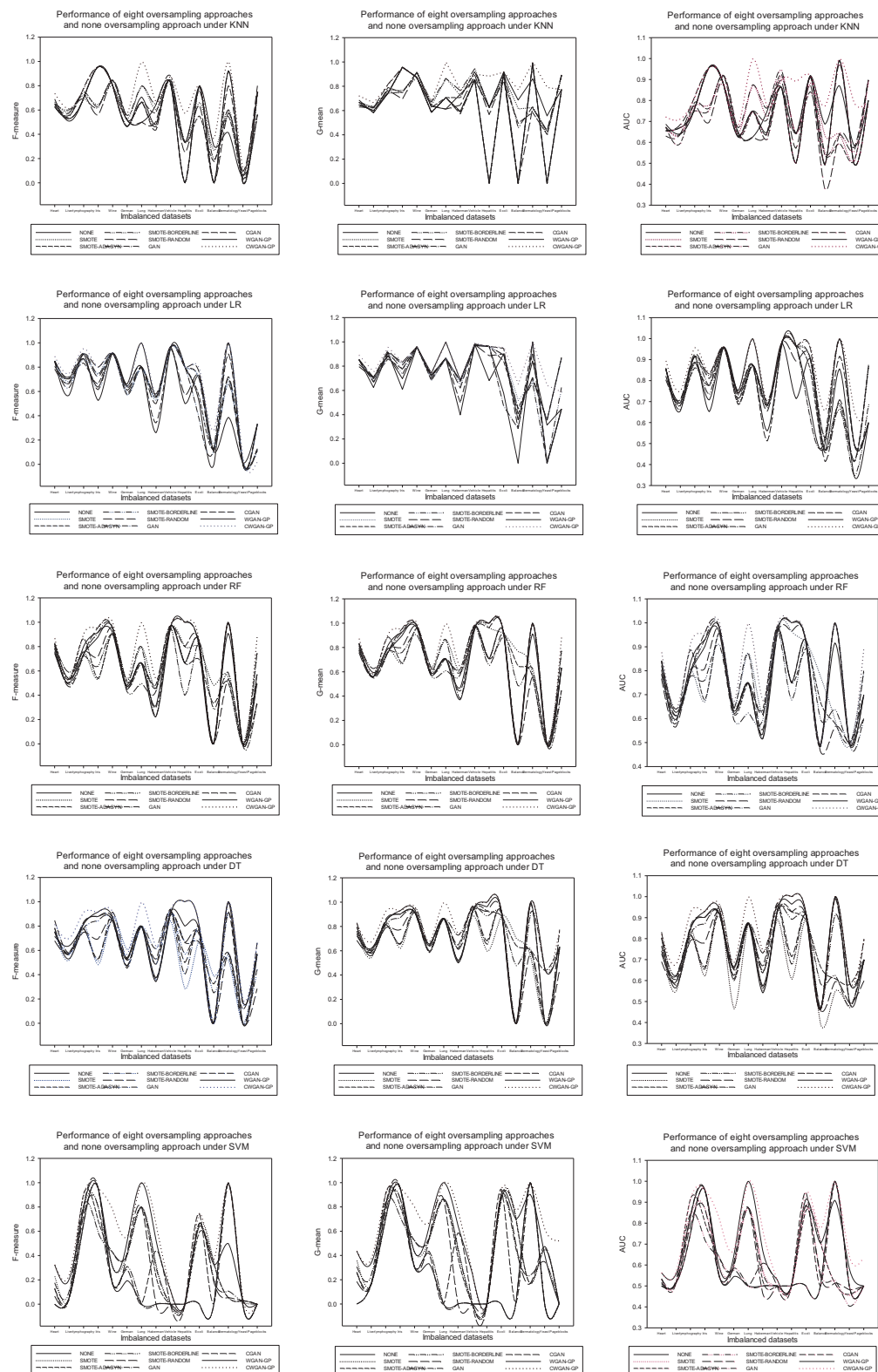


Fig. 4. Performance results of each of the oversampling approaches and the approach without oversampling using 15 different imbalanced datasets.

Table 4

Mean ranking score of eight oversampling approaches and approach without sampling under three evaluation metrics. Bold values indicate the best performing approach for each row.

Algorithms	Metrics	NONE	SMOTE	ADASYN	BORDERLINE SMOTE	RANDOM	GAN	CGAN	WGAN-GP	CWGAN-GP
KNN	F-measure	3.90	5.47	4.47	5.83	4.37	3.80	4.17	4.93	8.07
	G-mean	3.60	5.73	4.77	6.23	4.23	3.63	4.07	5.00	7.73
	AUC	3.90	5.27	4.63	5.70	3.93	4.03	4.33	5.47	7.73
LR	F-measure	3.93	4.97	4.73	4.20	4.67	4.97	5.00	4.93	7.60
	G-mean	3.47	5.20	4.97	4.37	5.33	4.03	4.53	5.03	8.07
	AUC	4.43	4.67	4.23	3.93	4.70	5.27	5.30	4.60	7.87
RF	F-measure	4.47	5.37	4.57	4.20	4.43	4.57	4.87	4.80	7.73
	G-mean	4.47	5.47	4.77	4.27	4.17	4.50	4.67	4.90	7.80
	AUC	4.57	4.83	4.30	3.97	3.77	4.80	5.40	5.03	8.33
DT	F-measure	6.03	3.77	4.00	4.60	3.17	5.07	5.47	4.70	8.20
	G-mean	6.17	3.30	3.93	5.07	3.17	5.00	5.27	4.83	8.27
	AUC	6.27	2.63	4.07	5.07	3.23	5.10	5.57	4.90	8.17
SVM	F-measure	3.17	5.17	4.80	5.03	4.10	3.30	4.47	6.63	8.33
	G-mean	3.17	5.17	4.93	4.90	4.10	3.30	4.47	6.63	8.33
	AUC	3.47	4.77	4.73	4.57	3.83	3.60	4.87	6.83	8.33

Table 5

Friedman test results.

Algorithms	Metrics	p-value	Algorithms	Metrics	p-value	Algorithms	Metrics	p-value
KNN	F-measure	1.13E–04	DT	F-measure	3.54E–06	RF	F-measure	0.0057
	G-mean	6.74E–05		G-mean	5.80E–07		G-mean	0.0029
	AUC	8.02E–04		AUC	1.47E–07		AUC	1.24E–04
LR	F-measure	0.0073	SVM	F-measure	7.16E–09			
	G-mean	8.42E–05		G-mean	7.38E–09			
	AUC	0.0012		AUC	1.70E–08			

the following: (1) the CWGAN-GP oversampling approach, represented by the red dotted line, outperformed the traditional oversampling approaches under three evaluation metrics; (2) compared with GAN and CGAN, CWGAN-GP showed superior performance for most classification algorithms and datasets; (3) similarly, the performance of CWGAN-GP was superior to that of WGAN-GP for most datasets with all classification algorithms under all metrics. Therefore, compared with the other oversampling approaches under all metrics, CWGAN-GP performed the best. In addition, the results of the performance evaluation were statistically confirmed by statistical tests.

4.2. Statistical test results

The mean ranking scores of all datasets under the three evaluation metrics are listed in Table 4. To determine whether these oversampling approaches exhibit the same performance, the Friedman test was applied; the results are shown in Table 5. These results indicate that at the significance level of $\alpha = 0.05$, the hypothesis that all oversampling approaches perform similarly in the mean ranking score for any evaluation metric is rejected (i.e., the performances of the oversampling approaches are significantly different). The Nemenyi post hoc test was used to distinguish the performances of each oversampling approach; the results are shown in Fig. 5.

Fig. 5 reveals that CWGAN-GP outperforms the other oversampling approaches. Moreover, when CWGAN-GP is used as the control method, the worst case only occurred when the classifier was LR and the evaluation metric was the F-measure. In this case, CWGAN-GP is significantly more effective than the two oversampling methods but not significantly different from other oversampling methods. In the best case, CWGAN-GP does not differ significantly from one of the oversampling approaches, but its performance is significantly superior to that of all other oversampling approaches; the best case occurs four times. Table 6 compares the significance of the oversampling approaches in detail.

4.3. Performance evaluation results on real imbalanced datasets

To further assess the generalization ability of CWGAN-GP and to verify its effectiveness for practical applications, we repeated the above experiments using two real imbalanced datasets from different fields. Fig. 6 shows the performance evaluation results of the F-measure, G-mean, and AUC for the eight oversampling approaches and the approach without sampling using the imbalanced traffic accident and software defect datasets. Clearly, CWGAN-GP exhibited better performance than the other oversampling approaches and the approach without sampling when applied to the imbalanced traffic accident dataset. In this case, the G-mean and AUC of CWGAN-GP achieved superior results for five classifiers. When the F-measure was used as the evaluation metric, CWGAN-GP also achieved superior performance results, except with the classifier LR, where the performance was the second best. On the imbalanced software defect dataset, using the classifiers KNN,

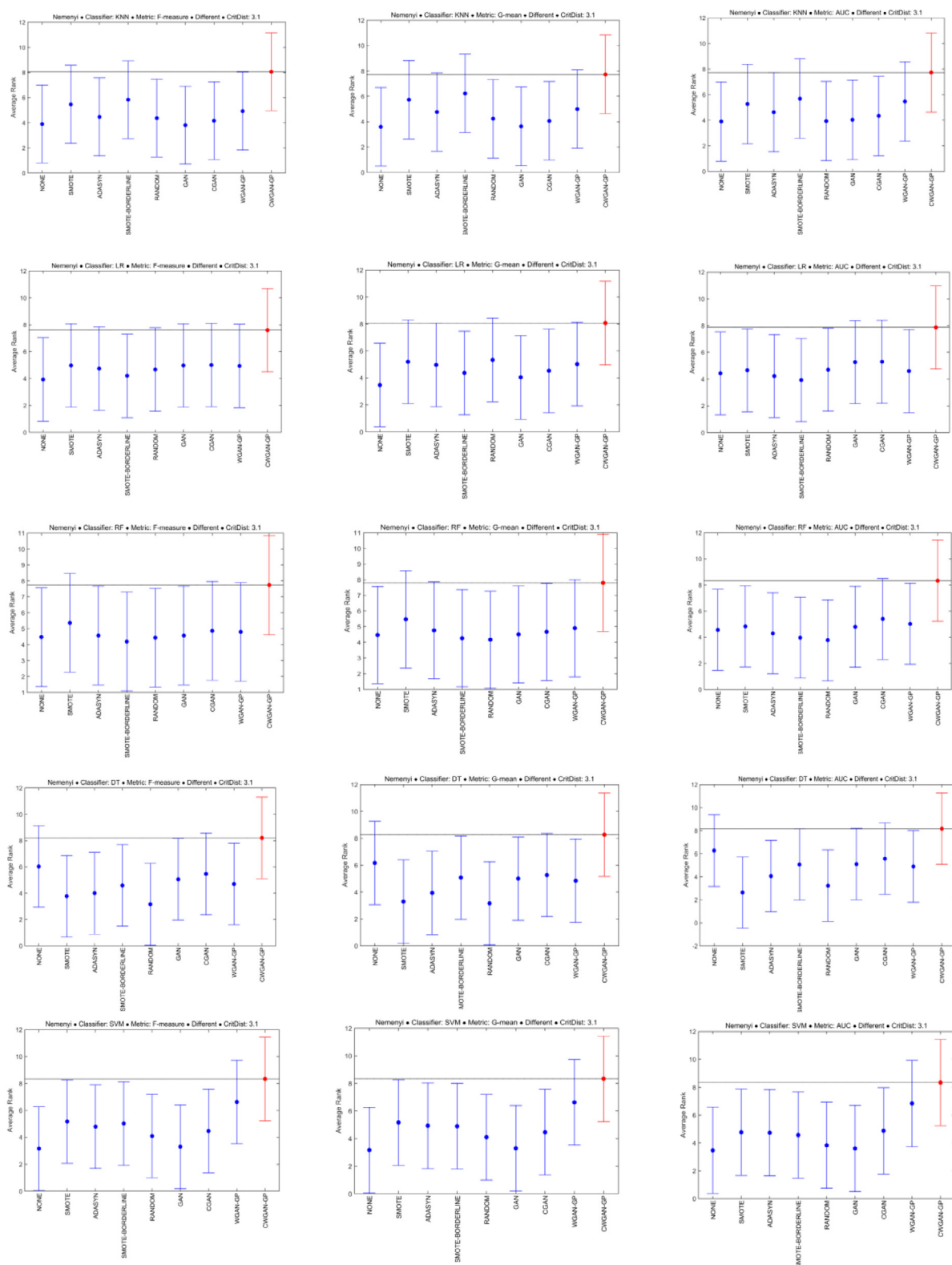
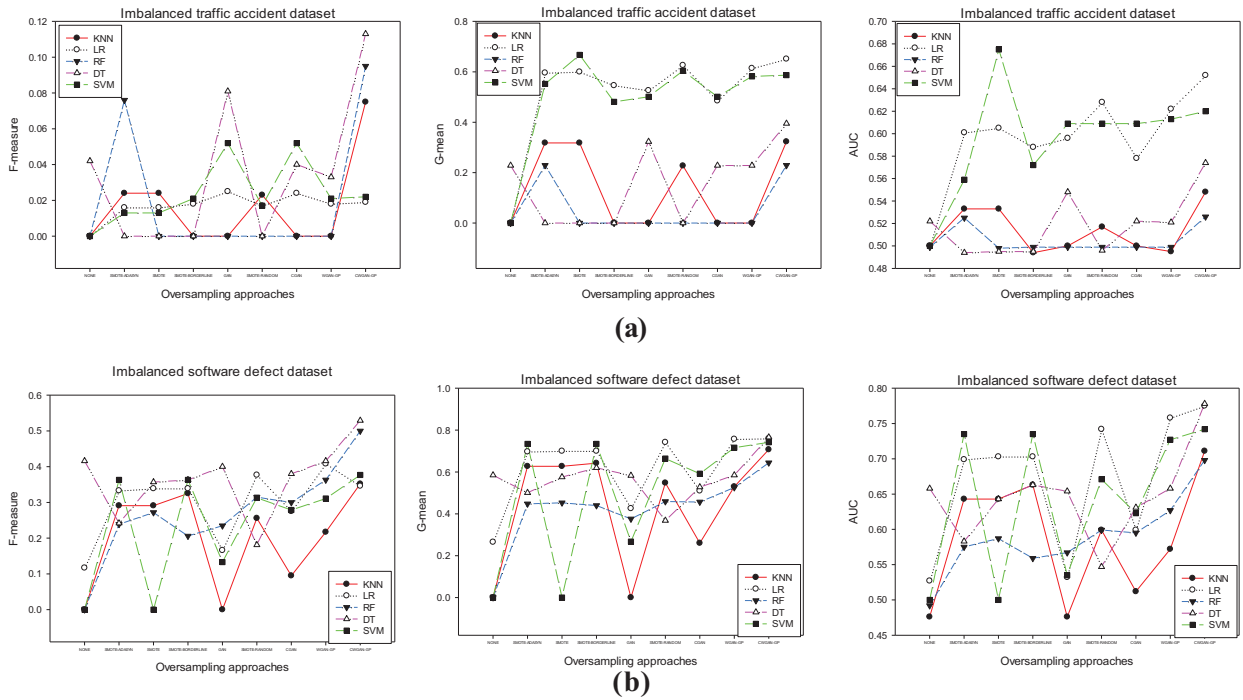


Fig. 5. Results of the Nemenyi post hoc test.

Table 6

Results of the Nemenyi post hoc test when CWGAN-GP is used as the control method.

Algorithm	Metric	Not significantly different from CWGAN-GP	Significantly different from CWGAN-GP
KNN	F-measure	SMOTE, BORDERLINE-SMOTE	All other
	G-mean	SMOTE, ADASYN, BORDERLINE-SMOTE, WGAN-GP	All other
	AUC	SMOTE, BORDERLINE-SMOTE, WGAN-GP	All other
LR	F-measure	SMOTE, ADASYN, RANDOM, GAN, CGAN, WGAN-GP	All other
	G-mean	SMOTE, RANDOM, WGAN-GP	All other
	AUC	GAN, CGAN	All other
RF	F-measure	SMOTE, CGAN, WGAN-GP	All other
	G-mean	SMOTE, ADASYN, WGAN-GP	All other
	AUC	CGAN	All other
DT	F-measure	NONE, CGAN	All other
	G-mean	NONE, CGAN	All other
	AUC	NONE, GAN, CGAN	All other
SVM	F-measure	WGAN-GP	All other
	G-mean	WGAN-GP	All other
	AUC	WGAN-GP	All other

**Fig. 6.** Performance each oversampling approach and the approach without oversampling when applied to two real imbalanced datasets: (a) imbalanced traffic accident dataset; (b) imbalanced software defect dataset.

RF, and DT, CWGAN-GP outperformed other oversampling approaches in terms of the F-measure, G-mean, and AUC when using LR as the classifier. The performance of CWGAN-GP was also superior to those of the other oversampling approaches for both the G-mean and AUC. The only exception occurred when using the SVM classifier, in which case the performance of CWGAN-GP was the third best for both the F-measure and G-mean and the second best for AUC.

4.4. Relationship between performance variation of oversampling approaches and IR

The results in Fig. 4 show that the performance of the oversampling approaches did not exhibit regular behavior with the IR for the three evaluation metrics. Among the five classification algorithms, the best and worst performance scores of all the oversampling approaches are 1 and 0, respectively, and the IR varies considerably for the imbalanced datasets. As a result, it is not possible to conclude that smaller or larger IR leads to better or worse performance. Fig. 4 suggests that the performance of the oversampling approaches may be relevant to the distribution of the data (i.e., the more complex the data distribution, the less the oversampling approach can learn about the characteristics of the data distribution); thus, the generated data will be less real, which worsens the performance. In contrast, the simpler the data distribution, the more

characteristics the oversampling approaches can learn from the data distribution. As such, the generated data will be more realistic and the performance will be better. This is supported by the fact that all of the oversampling approaches show similar performance variations for the same imbalanced dataset.

4.5. Explaining the performance of CWGAN-GP

As shown in Fig. 4, the good performance of CWGAN-GP can mainly be attributed to the following three advantages.

- (1) Traditional oversampling approaches pay more attention to local data information, whereas our data generation model is more concerned with the true distribution of data and directly generates synthetic data.
- (2) CWGAN-GP outperforms GAN and CGAN because it uses the EM distance as the measure of the distance between the true data distribution and the generative data distribution; in contrast, GAN and CGAN use JS and KL divergence. The different ways to measure distance between real and generative data distributions are responsible for the problems of mode collapse and unstable training during the training process of GAN and CGAN, but this is rarely an issue during the training of CWGAN-GP.
- (3) Similarly, CWGAN-GP outperforms WGAN-GP because of the addition of auxiliary conditional information.

4.6. Limitation of CWGAN-GP compared with traditional approaches

Training models based on GAN are known to have limitations [21]; the limitation of our approach is that CWGAN-GP requires more time and effort for training than traditional oversampling approaches. However, in most scenarios, exchanging limited time and effort for higher quality synthetic data is worthwhile as it considerably improves the performance of the classification algorithms.

4.7. Challenges of CWGAN-GP

Both theoretical and applied research into GAN-based oversampling to alleviate imbalanced data classification are developing rapidly. The main challenges facing the CWGAN-GP oversampling approach proposed in this work are as follows.

- (1) Compared with GAN-based oversampling approaches, CWGAN-GP can better address training instability to some extent. However, as the stability of data generation must be investigated by using different designs under different network architectures, this problem is still worth studying.
- (2) As the training process of CWGAN-GP consumes more time and energy than do traditional oversampling approaches, theoretical research into CWGAN-GP and accelerating its training and convergence processes are required.
- (3) Although all oversampling approaches can generate the required data, the quality and diversity of the generated data differ from those of real data. To further increase the quality and diversity of data generated in various application fields, researchers must develop new approaches and propose more effective methods.

5. Related works

Our work is relevant to two research axes: (1) using oversampling approaches to alleviate the problem of imbalanced data classification; and (2) using GANs and extensions thereof to alleviate the imbalanced data classification problem.

5.1. Oversampling approaches to alleviate imbalanced data classification

Oversampling approaches mainly focus on balancing the distributions of majority and minority classes by increasing the number of samples in the minority class. This strategy involves analyzing the minority class and generating new synthetic samples, which are then added to the training dataset. The simplest oversampling approach is ROS, which randomly duplicates the samples of the minority class to generate new samples, increasing the size of the minority class. However, this can result in overfitting because it provides no new information to the classifier in the course of training.

SMOTE, an improved method proposed by Chawla et al. [14], mainly generates synthetic samples along the line segment that joins the minority class. Various modifications of SMOTE have been proposed; for example, Smote ++ Tomek and Smote ++ Edited Nearest Neighbor were proposed to overcome a problem related to using an excessively small minority class for training in the presence of other complicating factors, such as class overlapping. On the basis of SMOTE, Han et al. [28] proposed borderline-SMOTE1 and borderline-SMOTE2, in which only the samples of the minority class near the borderline are oversampled because the data on the borderline of the classification boundary are the most likely to be misclassified. An ADASYN sampling approach was also presented, which uses a weighted distribution for different samples of the minority class according to their level of difficulty in learning [28]. This approach generates more synthetic samples for the minority class that are harder to learn than samples that are easier to learn. The Majority Weighted Minority Oversampling Technique (MWMOTE) first identifies the samples of the minority class and assigns them weights according to their Euclidean distance from the nearest samples of the majority class; then, it generates synthetic samples from the weighted samples of the minority class using a clustering method [29].

Table 7
Comparison of GAN and its extensions as an oversampling approach

	AC	ST	MC
GAN	–	–	–
CGAN	+	–	–
WGAN-GP	–	+	+
Our approach	+	+	+

Bennin et al. [49] pointed out that these approaches shared similar procedures for generating synthetic data; that is, they all use the k -nearest neighbor (k -NN) technique to select similar neighbors for oversampling. When the samples of the minority class contain several sub-clusters of unequal sizes (an intra-class imbalance), these k -NN-based approaches are still deficient and are sometimes inappropriate for certain disparate data classification issues.

5.2. GAN-based methods to alleviate imbalanced data classification

In recent years, the field of artificial intelligence has progressed significantly, especially in the realm of machine learning. GAN, which was developed on the basis of a depth generation model and differs from k -NN-based approaches, is no longer based on the local information of an imbalanced dataset but captures the true distribution of data and directly generates synthetic data. To alleviate the imbalanced data classification problem, some researchers have focused on GAN and its extensions as a new oversampling approach. Fiore et al. [24] trained a GAN to output mimicked minority class samples, which were then combined with training data into an augmented training set. Their experiments revealed that the effectiveness of a classifier can be improved in the specific application domain of credit card fraud detection. Douzas et al. [16] introduced a CGAN-based network that captures the true distribution of the global information of the minority class by adding additional conditional information to GAN, and thereafter generates synthetic data for the minority class in an imbalanced dataset. Their empirical results revealed that generative data is significantly improved when CGAN is used as an oversampling algorithm, although this method is sometimes problematic as it experiences unstable training and mode collapse.

A comparative summary of previous efforts in this area is provided in Table 7. The columns of the table correspond to the following criteria.

- **AC** indicates whether the approach involves auxiliary condition information: existence (+), nonexistence (–).
- **ST** indicates whether the training is stable in this approach: stable (+), unstable (–).
- **MC** indicates whether mode collapse is likely to occur when this method is used as the oversampling method: existence (–), nonexistence (+).

6. Conclusions

In both theoretical research and real-world applications, imbalanced data are a widespread phenomenon. Traditional classification algorithms are applied based on relatively balanced distributions of data classes. However, these algorithms may have different degrees of defects and may thus become inefficient. To alleviate the problem caused by imbalanced data classification, resampling approaches have been proposed, the majority of which use oversampling techniques. Oversampling is based on the principle of generating minority data to balance an imbalanced dataset. Although these oversampling approaches have been shown to increase performance, traditional oversampling approaches focus on the local information of the imbalanced dataset, such that the generated data is insufficiently realistic. Furthermore, oversampling approaches based on GAN always experience the problems of mode collapse and unstable training. To address these problems, we proposed CWGAN-GP, a novel oversampling approach based on WGAN-GP and which is motivated by the conditional version of GAN. Our approach generates new data by learning the distribution characteristics of the real data.

We empirically evaluated our oversampling approach CWGAN-GP by comparing it with eight other oversampling approaches (SMOTE, ADASYN, BORDERLINE-SMOTE, RANDOM, GAN, CGAN, and WGAN-GP) and an approach without sampling. We used five classification algorithms (KNN, LR, RF, DT, and SVM) on 15 different benchmarked imbalanced datasets and two real imbalanced datasets. The experimental results show that CWGAN-GP outperformed the other oversampling approaches under all metrics; statistical tests confirm the validity of the experimental results.

Three possible explanations exist for the relatively superior performance of CWGAN-GP. First, compared with traditional oversampling approaches, which generate synthetic data along the line segment that joins minority class data and pay more attention to local data information, CWGAN-GP is more concerned with the true distribution of data and directly generates synthetic data. Second, whereas CWGAN-GP uses the EM distance, GAN and CGAN use the JS divergence, which causes mode collapse and unstable training. In contrast, these problems rarely occur when training CWGAN-GP. Third, CWGAN-GP outperforms WGAN-GP because CWGAN-GP uses conditional information for auxiliary training, unlike WGAN-GP. In future studies, we intend to apply the CWGAN-GP oversampling approach to other real-world imbalanced data classification problems pertaining to specific practical applications. We also aim to compare the performance of CWGAN-GP to that of complex oversampling approaches.

Declaration of Competing Interest

This manuscript has not been published or presented elsewhere in part or in entirety and is not under consideration by another journal. We have read and understood your journal's policies, and we believe that neither the manuscript nor the study violates any of these. All authors have approved the final article. There are no conflicts of interest to declare.

Acknowledgments

This work was supported by the [National Natural Science Foundation of China](#) (No. 61662085, No. 61862065), the Yunnan Province Ph.D. Scholar Newcomer Award, the second round of the "Yunnan University Service Yunnan Action Plan" project, the research and application of big data of intelligent transportation in Yunnan Province ([2016ZD05](#)), the Yunnan Provincial Natural Science Foundation Fundamental Research Project ([2019FB-16](#)), The Project of Yunnan Provincial Department of Education Science Research Fund ([2017ZZX227](#)), the Yunnan University Data-Driven Software Engineering Provincial Science and Technology Innovation Team Project ([2017HC012](#)), the Yunnan University "Dong Lu Young-backbone Teacher" Training Program and the Yunnan University Education Department Science Research Fund Graduate Program ([2019Y0008](#), [2019Y0010](#), respectively).

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